



DIGITAL LOGIC DESIGN (DLD)

By: Dr. Syed Shahid Sultan

Chapter 2: . Boolean Algebra and Logic Gates

Week 2: Lecture 4



Reference Readings and Acknowledgements

Chapter 2: Digital Design with Introduction to
Verilog HDL M. Morris Mano, Michael D.
Ciletti, 5th Edition



Contents

- Basic Definitions
- Axiomatic Definition of Boolean Algebra
- Two-valued Boolean Algebra
- Basic Theorems and Properties
- Boolean Functions
- Algebraic Manipulation
- Complement of a Function
- Canonical and Standard Forms
- Digital Logic Gates
- Extension to Multiple Inputs



Basic Definitions

- A **set** of elements is any collection of objects, usually having a common property.
- If S is a set, and x and y are certain objects,
 - $x \in S$ means that x is a member of the set S .
 - $y \notin S$ means that y is not an element of the set S .
- A binary operator defined on a set S of elements is a rule that assigns, to each pair of elements from S , a unique element from S .
 - $a*b=c$
 - $*$ is a binary operator if it specifies a rule for finding c from the pair (a,b) and also if $a, b, c \in S$.
 - $*$ is not a binary operator if $a, b \in S, c \notin S$.



Basic Definitions (cont.)

The Postulates of a mathematical system form the basic assumptions from which it is possible to deduce the rules, theorems, and properties of the system. The most common postulates used to formulate various structures are as follows

1. Closure

- A set S is closed with respect to a binary operator if, for every pair of elements of S , the binary operators specifies a rule for obtaining a unique element of S .
 - The set of natural number $N=\{1,2,3,4,\dots\}$ is closed with respect to the binary operator $+$ by the rules of arithmetic addition, since, for any $a, b \in N$, there is a unique $c \in N$ such that $a+b=c$.
 - The set of natural number N is not closed with respect to the binary operator $-$ by the rules of arithmetic subtraction, since, $2-3=-1$, $2,3 \in N$, but $(-1) \notin N$.



Basic Definitions (cont.)

2. Associative law

- A binary operator $*$ on a set S is said to be associative whenever

$$(x * y) * z = x * (y * z) \text{ for all } x, y, z \in S$$

3. Commutative law

- A binary operator $*$ on a set S is said to be commutative whenever

$$x * y = y * x \text{ for all } x, y \in S$$

4. Identity element

- A set S is said to have an identity element with respect to a binary operation $*$ on S if there exists an element $e \in S$ with the property that

$$e * x = x * e = x \text{ for every } x \in S$$



Basic Definitions (cont.)

5. Inverse

- A set S having the identity element e with respect to a binary operator $*$ is said to have an inverse whenever, for every $x \in S$, there exists an element $y \in S$ such that

$$x * y = e$$

- Example : In the set of integers, I , and the operator $+$, with $e=0$, the inverse of an element a is $(-a)$, since $a + (-a) = 0$.

6. Distributive law

- IF $*$ and $\cdot$ are two binary operators on a set S , $*$ is said to be distributive over $\cdot$ whenever

$$x * (y \cdot z) = (x * y) \cdot (x * z)$$



Axiomatic Definition of Boolean Algebra

Boolean Algebra is an algebraic structure defined by a set of element, B , together with two operators, $+$ and $\cdot$, provided that following Huntington postulates are satisfied:

1. Closure with respect to the operator $+$ ($\cdot$)

- $x, y \in B$ such that $x+y \in B$

2. An identity element w.r.t. $+$ ($\cdot$)

- $0+x = x+0 = x$
- $1 \cdot x = x \cdot 1 = x$

3. Commutative w.r.t. $+$ ($\cdot$)

- $x+y = y+x$
- $x \cdot y = y \cdot x$



Axiomatic Definition of Boolean Algebra (cont.)

4. $\cdot$ is distributive over $+$: $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$
 $+$ is distributive over $\cdot$: $x + (y \cdot z) = (x + y) \cdot (x + z)$

$\forall$ for every element (for all)
 $\exists$ there exists

5. $\forall x \in B, \exists x' \in B$ (complement of x)
such that $x + x' = 1$ and $x \cdot x' = 0$

6. $\exists$ at least two elements $x, y \in B$ such that $x \neq y$

Comparing Boolean algebra with arithmetic algebra, we note the following differences:

- Huntington postulates do not include the associative law, however, it can be derived from other postulates.
- The distributive law of $+$ over $\cdot$ ($x + (y \cdot z) = (x + y) \cdot (x + z)$), is valid for Boolean algebra, but not for arithmetic algebra.
- Boolean algebra does not have additive and multiplicative inverses
- Complement is not available in ordinary algebra.
- Boolean algebra deal with two-valued Boolean number. Ordinary algebra deals with the real number.



Two-valued Boolean Algebra

A two valued Boolean algebra is defined on a set of two elements, $B = \{0,1\}$, with rules for the operations $+$ and $\cdot$.

x	y	$x \cdot y$	x	y	$x + y$	x	x'
0	0	0	0	0	0	0	1
0	1	0	0	1	1	1	0
1	0	0	1	0	1		
1	1	1	1	1	1		

The Huntington postulates are valid for the set B and the two binary operator $+$ and $\cdot$.

- Closure
 - The identity elements
- a) $0+0=0$, $0+1=1+0=1$;
b) $1 \cdot 1=1$, $1 \cdot 0=0 \cdot 1=0$.

Two-valued Boolean Algebra (cont.)

- The commutative laws are obvious from the symmetry of the binary operator tables.
- The distributive laws
 - $x(y+z) = xy + xz$ can be shown to hold from the operator tables.

x	y	z	$y + z$	$x \cdot (y + z)$	$x \cdot y$	$x \cdot z$	$(x \cdot y) + (x \cdot z)$
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1



Two-valued Boolean Algebra (cont.)

- Complement
 - $x+x'=1$: $0+0'=0+1=1$; $1+1'=1+0=1$
 - $x \cdot x'=0$: $0 \cdot 0'=0 \cdot 1=0$; $1 \cdot 1'=1 \cdot 0=0$
- Postulate 6 is satisfied because the two-valued Boolean Algebra has two distinct elements 1 and 0, with $0 \neq 1$

Note

- a set of two elements
- $+$: OR operation; $\cdot$: AND operation
- a complement operator: NOT operation
- Binary logic is a two-valued Boolean algebra



Basic Theorems and Properties

Duality

- the binary operators are interchanged; AND $\Leftrightarrow$ OR
- the identity elements are interchanged; 1 $\Leftrightarrow$ 0

Table 2.1

Postulates and Theorems of Boolean Algebra

Postulate 2	(a) $x + 0 = x$	(b) $x \cdot 1 = x$
Postulate 5	(a) $x + x' = 1$	(b) $x \cdot x' = 0$
Theorem 1	(a) $x + x = x$	(b) $x \cdot x = x$
Theorem 2	(a) $x + 1 = 1$	(b) $x \cdot 0 = 0$
Theorem 3, involution	$(x')' = x$	
Postulate 3, commutative	(a) $x + y = y + x$	(b) $xy = yx$
Theorem 4, associative	(a) $x + (y + z) = (x + y) + z$	(b) $x(yz) = (xy)z$
Postulate 4, distributive	(a) $x(y + z) = xy + xz$	(b) $x + yz = (x + y)(x + z)$
Theorem 5, DeMorgan	(a) $(x + y)' = x'y'$	(b) $(xy)' = x' + y'$
Theorem 6, absorption	(a) $x + xy = x$	(b) $x(x + y) = x$



Basic Theorems and Properties

Theorem 1(a): $x+x = x$

$x+x = (x+x) 1$	by postulate:	2(b)
$= (x+x) (x+x')$		5(a)
$= x+xx'$		4(b)
$= x+0$		5(b)
$= x$		2(a)

– Theorem 1(b): $x x = x$

$$\begin{aligned} - \quad & xx = x x + 0 \\ & = xx + xx' \\ & = x (x + x') \\ & = x 1 \\ & = x \end{aligned}$$



Basic Theorems and Properties

■ Theorem 2

- $$\begin{aligned}x + 1 &= 1 (x + 1) \\&= (x + x')(x + 1) \\&= x + x' 1 \\&= x + x' \\&= 1\end{aligned}$$

- $x 0 = 0$ by duality

■ Theorem 3: $(x')' = x$

- Postulate 5 defines the complement of x , $x + x' = 1$ and $x x' = 0$
- The complement of x' is x is also $(x')'$

Basic Theorems and Properties

- Theorem 6

- $x + xy = x.1 + xy$
 $= x(1 + y)$
 $= x.1$
 $= x$

- $x(x + y) = x$ by duality

- By means of truth table

x	y	xy	$x + xy$
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1



Basic Theorems and Properties

DeMorgan's Theorems

- $(x+y)' = x' y'$
- $(x y)' = x' + y'$

x	y	$x+y$	$(x+y)'$	x'	y'	$x'y'$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0



Basic Theorems and Properties

- Operator Precedence
 - parentheses
 - NOT
 - AND
 - OR



Boolean Functions

A Boolean function is an algebra that deals with binary variables and logic operations.

- binary variables
- binary operators OR and AND
- unary operator NOT
- parentheses

Examples

- $F_1 = x y z'$
- $F_2 = x + y'z$
- $F_3 = x' y' z + x' y z + x y'$
- $F_4 = x y' + x' z$



Boolean Functions (cont.)

- The truth table of 2^n entries

$$F_1 = x y z'$$

$$F_2 = x + y'z$$

$$F_3 = x' y' z + x' y z + x y'$$

$$F_4 = x y' + x' z$$

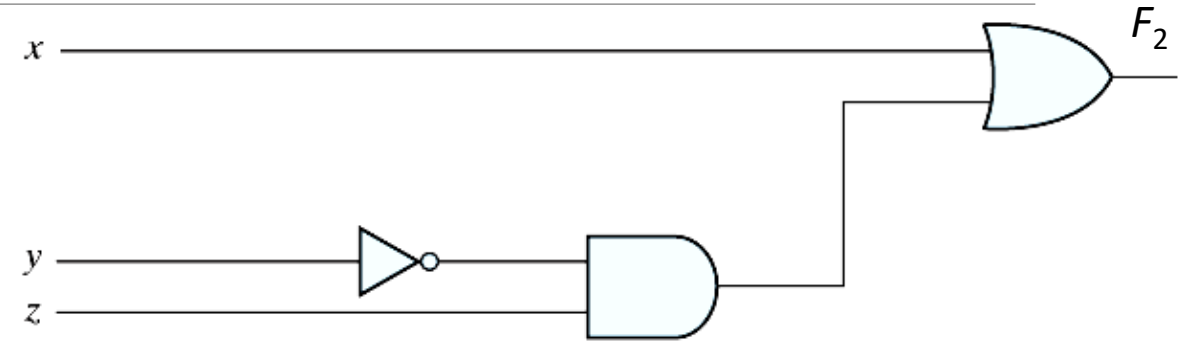
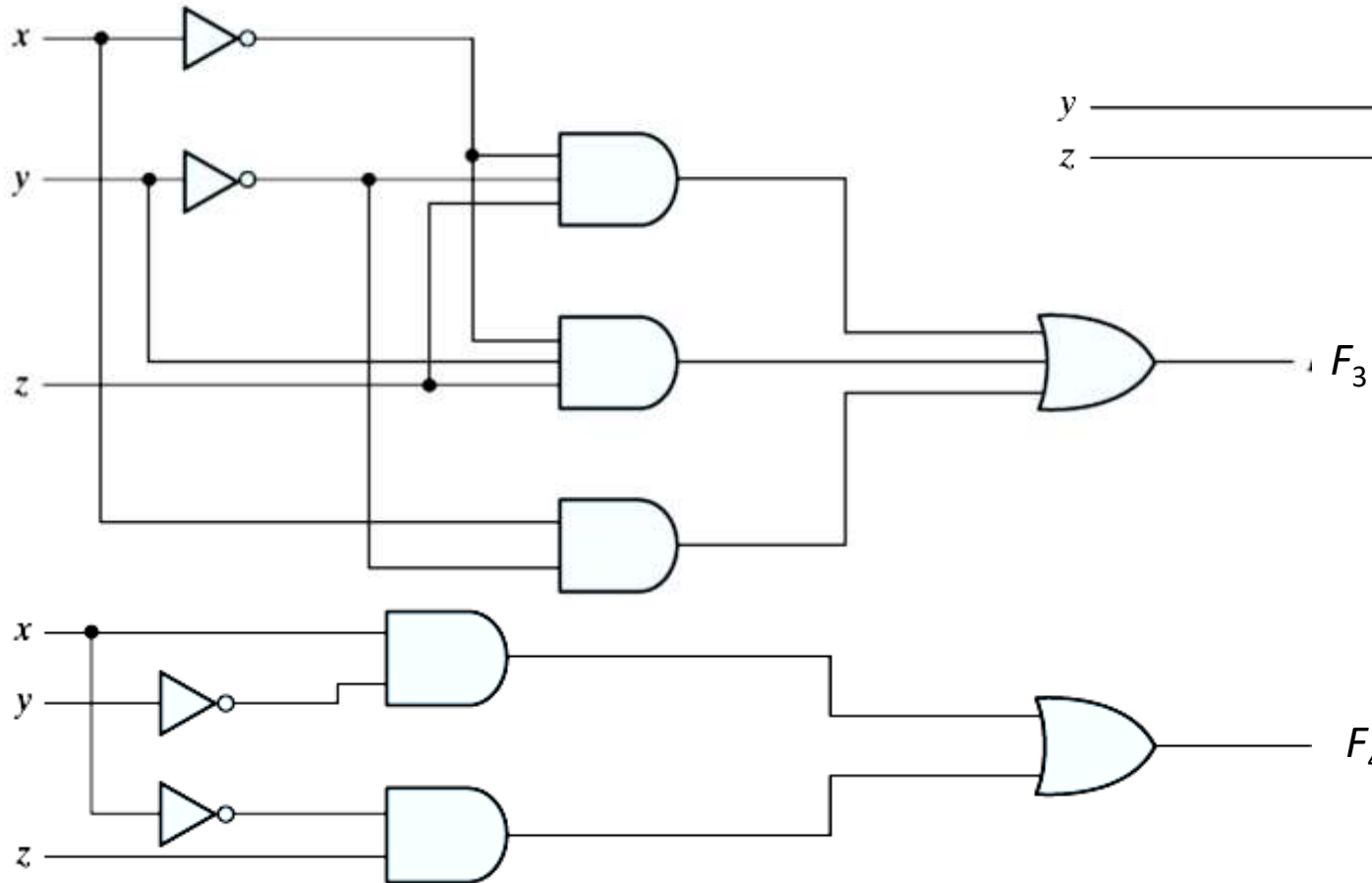
x	y	z	F_1	F_2	F_3	F_4
0	0	0	0	0	0	0
0	0	1	0	1	1	1
0	1	0	0	0	0	0
0	1	1	0	0	1	1
1	0	0	0	1	1	1
1	0	1	0	1	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	0

Two Boolean expressions may specify the same function

- $F_3 = F_4$

Boolean Functions (cont.)

- Implementation with logic gates
- F_4 is more economical



$$F_1 = x y z'$$

$$F_2 = x + y'z$$

$$F_3 = x'y'z + x'yz + xy'$$

$$F_4 = xy' + x'z$$